

Boundary Value Problems and Sturm–Liouville Theory

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Unit 17: Eigenfunctions, Orthogonality, and the Functional Basis

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Unit 17 at a Glance: From Side Conditions to a Functional Basis

An **initial** value problem pins a solution down with all its side data at one point; a **boundary** value problem spreads that data across two endpoints, and existence and uniqueness are no longer guaranteed. The interesting case is the homogeneous eigenvalue BVP $X'' + \lambda X = 0$, which is silent (only $X \equiv 0$) except at a discrete ladder of **eigenvalues** λ_n , where it rings with **eigenfunctions** X_n . Those eigenfunctions are mutually *orthogonal*, and orthogonality is the master key: it turns the infinite expansion $f = \sum c_n y_n$ into a list of one-line coefficient integrals. The **Sturm–Liouville** form $(p y')' + (q + \lambda w)y = 0$ packages every such problem and guarantees the eigenfunctions form a complete, weighted-orthogonal basis.

- **IVP vs. BVP** — moving the side conditions to two endpoints can make a linear problem have *no* solution, *one*, or *infinitely many*.
- **Eigenvalues / eigenfunctions** — $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ is nontrivial only at $\lambda_n = (n\pi/L)^2$ with $X_n = \sin \frac{n\pi x}{L}$.
- **The shooting method** — recast a BVP as an IVP with a *guessed* initial slope, then adjust the slope until the far boundary condition is met.
- **Orthogonality** — $\int_a^b y_n y_m w \, dx = 0$ for $n \neq m$ makes each coefficient $c_n = \langle f, y_n \rangle / \langle y_n, y_n \rangle$ an independent integral.
- **Sturm–Liouville** — $(p y')' + (q + \lambda w)y = 0$ has real eigenvalues $\lambda_n \rightarrow \infty$ and w -orthogonal eigenfunctions that form a complete basis $f = \sum c_n y_n$.

17.1 Initial versus Boundary Value Problems

Side conditions at one point (IVP) versus two endpoints (BVP); Dirichlet (y), Neumann (y'), and Robin conditions; the loss of the existence–uniqueness guarantee; steady-state heat conduction as the model BVP.

IVP versus BVP — Where the Data Lives

IVP: $y'' = F(x, y, y')$, $y(x_0) = y_0$, $y'(x_0) = y'_0$ (all data at one point x_0); **BVP:** $y(a) = \alpha$, $y(b) = \beta$ (d

The Three Standard Boundary Conditions

Dirichlet: $y(a) = \alpha$ (fixes the value); Neumann: $y'(a) = \alpha$ (fixes the slope/flux); Robin: $c_1 y(a) + c_2 y'(a) = \alpha$

Existence and Uniqueness — The Contrast

A linear IVP (continuous coefficients) $\Rightarrow$ **exactly one** solution; a linear BVP $\Rightarrow$ **none, one, or infinitely**

Method — Solve a Linear Two-Point BVP

1. Find the general solution of the ODE, carrying both arbitrary constants c_1, c_2 .
2. Impose the condition at the left endpoint $x = a$ to get one equation in c_1, c_2 .
3. Impose the condition at the right endpoint $x = b$ to get a second equation.
4. Solve the 2×2 system: a unique (c_1, c_2) gives one solution; an inconsistent system gives none; a dependent system gives infinitely many.

Worked Example — A Simple Dirichlet BVP

Solve $y'' = 0$ on $[0, 2]$ with $y(0) = 1$ and $y(2) = 5$.

General solution: $y'' = 0 \Rightarrow y(x) = c_1 + c_2 x$.

Left BC: $y(0) = c_1 = 1 \Rightarrow c_1 = 1$.

Right BC: $y(2) = c_1 + 2c_2 = 5 \Rightarrow 1 + 2c_2 = 5 \Rightarrow c_2 = 2$.

Unique solution: $y(x) = 1 + 2x$ (a single straight line meeting both endpoints).

Key Takeaway

An initial value problem stacks all its side data at **one** point and is rewarded with a unique solution; a boundary value problem spreads the data across **two** endpoints and forfeits that guarantee. The same ODE can then have no solution, one, or infinitely many — which is exactly what makes BVPs the natural home for eigenvalues.

Common Pitfall

Do not carry the IVP intuition — “a second-order equation always has a unique solution given two conditions” — into BVPs. **Two boundary conditions need not pin down a unique solution.** Whether you get none, one, or infinitely many depends on how the conditions interact with the general solution, not merely on counting constants.

17.2 Two-Point Boundary Value Problems

The standard solve-then-impose method for linear two-point BVPs; the three possible outcomes; homogeneous BVPs and the omnipresent trivial solution; nonhomogeneous BVPs with forcing.

The Linear Two-Point BVP

$$y'' + P(x)y' + Q(x)y = R(x), \quad y(a) = \alpha, \quad y(b) = \beta; \quad y = c_1 y_1 + c_2 y_2 + y_p \text{ before imposing the BCs.}$$

Three Possible Outcomes

$$\det \begin{pmatrix} y_1(a) & y_2(a) \\ y_1(b) & y_2(b) \end{pmatrix} \neq 0 \Rightarrow \textbf{unique}; \quad = 0 \Rightarrow \textbf{none or } \infty\text{-many (depending on } \alpha, \beta).$$

Homogeneous BVP — The Trivial Solution

$$y'' + \lambda y = 0, \quad y(0) = y(L) = 0 \Rightarrow y \equiv 0 \text{ always solves it; nontrivial } y \text{ exist only at special } \lambda.$$

Method — Resolve a Two-Point BVP

1. Solve the ODE completely, including a particular solution y_p if the equation is nonhomogeneous.
2. Substitute the two boundary conditions to obtain a linear system in c_1, c_2 .
3. Compute the boundary determinant: nonzero means a unique solution exists — solve for c_1, c_2 .
4. If the determinant is zero, test consistency: an impossible equation gives no solution, a redundant one gives a one-parameter family of solutions.

Worked Example — A Nonhomogeneous BVP

Solve $y'' = 2$ on $[0, 1]$ with $y(0) = 0$ and $y(1) = 0$.

$$\textbf{General solution: } y'' = 2 \Rightarrow y' = 2x + c_2 \Rightarrow y(x) = x^2 + c_2 x + c_1.$$

$$\textbf{Left BC: } y(0) = c_1 = 0 \Rightarrow c_1 = 0.$$

$$\textbf{Right BC: } y(1) = 1 + c_2 + 0 = 0 \Rightarrow c_2 = -1.$$

$$\textbf{Unique solution: } y(x) = x^2 - x \text{ (a downward parabola pinned to 0 at both ends).}$$

Key Takeaway

Every linear two-point BVP is solved the same way: find the **general solution first**, then let the two boundary conditions decide the constants. The boundary determinant is the referee — nonzero awards a unique solution, while zero throws the problem into the no-solution or infinitely-many camp, the very fork that produces eigenvalues.

Common Pitfall

Never impose a boundary condition before you have the *full* general solution (including y_p). Dropping a homogeneous piece or the particular solution silently discards the freedom the BCs need, and you will report a unique answer where the true problem has none or infinitely many.

17.3 Eigenvalues and Eigenfunctions of Boundary Value Problems

The eigenvalue BVP $X'' + \lambda X = 0$ with homogeneous BCs; the three-case analysis $\lambda < 0$, $\lambda = 0$, $\lambda > 0$; Dirichlet eigenvalues $\lambda_n = (n\pi/L)^2$ with eigenfunctions $\sin(n\pi x/L)$; the Neumann variant with cosines.

The Eigenvalue Boundary Value Problem

$X'' + \lambda X = 0$, $X(0) = 0$, $X(L) = 0$; λ is an *eigenvalue* if a nontrivial X (the *eigenfunction*) exists.

Dirichlet Spectrum

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots \quad (\lambda_n \rightarrow \infty).$$

Neumann Spectrum

$$X'(0) = X'(L) = 0: \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \cos \frac{n\pi x}{L}, \quad n = 0, 1, 2, \dots \quad (\lambda_0 = 0 \text{ allowed}, X_0 = 1).$$

Method — Find the Eigenvalues and Eigenfunctions

1. Split into three cases by the sign of λ , since the form of the general solution differs for $\lambda < 0$, $\lambda = 0$, and $\lambda > 0$.
2. For $\lambda < 0$ (write $\lambda = -\mu^2$) and $\lambda = 0$, impose the BCs and show only the trivial solution survives — no negative or zero eigenvalues here.
3. For $\lambda > 0$ (write $\lambda = \mu^2$) use $X = A \cos \mu x + B \sin \mu x$; the left BC kills A , and the right BC forces $\sin \mu L = 0$.
4. Solve the quantization condition $\mu L = n\pi$ to read off $\lambda_n = (n\pi/L)^2$ and the eigenfunctions $X_n = \sin(n\pi x/L)$.

Worked Example — Full Three-Case Analysis on $0 \leq x \leq \pi$

Find all eigenvalues and eigenfunctions of $X'' + \lambda X = 0$ with $X(0) = 0$, $X(\pi) = 0$.

Case $\lambda < 0$ ($\lambda = -\mu^2$): $X = A \cosh \mu x + B \sinh \mu x$; $X(0) = A = 0$, $X(\pi) = B \sinh \mu \pi = 0 \Rightarrow B = 0$. Only $X \equiv 0$.

Case $\lambda = 0$: $X = A + Bx$; $X(0) = A = 0$, $X(\pi) = B\pi = 0 \Rightarrow B = 0$. Only $X \equiv 0$.

Case $\lambda > 0$ ($\lambda = \mu^2$): $X = A \cos \mu x + B \sin \mu x$; $X(0) = A = 0 \Rightarrow X = B \sin \mu x$.

Quantization: $X(\pi) = B \sin \mu \pi = 0$, $B \neq 0 \Rightarrow \sin \mu \pi = 0 \Rightarrow \mu \pi = n\pi \Rightarrow \mu = n$.

Spectrum: $\lambda_n = n^2$, $X_n(x) = \sin(nx)$, $n = 1, 2, 3, \dots$

Key Takeaway

A homogeneous BVP is a **filter**: it passes only the trivial solution except at a discrete ladder of eigenvalues $\lambda_n = (n\pi/L)^2$, where it resonates with the eigenfunctions $\sin(n\pi x/L)$. The negative and zero cases always collapse to $X \equiv 0$; the entire spectrum lives in $\lambda > 0$ and marches off to infinity.

Common Pitfall

You **must check all three sign cases** of λ — skipping the $\lambda \leq 0$ analysis is only safe once you have *shown* it yields nothing. Also, $n = 0$ gives $X \equiv 0$ for Dirichlet (so start at $n = 1$), but $n = 0$ **is** a genuine eigenvalue $\lambda_0 = 0$ with $X_0 = 1$ in the Neumann case.

17.4 The Shooting Method

Recasting a BVP as an IVP with a guessed initial slope s ; the boundary-mismatch function $F(s) = y(b; s) - \beta$; root-finding on F ; for linear BVPs, exactly two shots plus linear interpolation; reuse of Runge–Kutta integrators.

BVP Recast as a Parameterized IVP

$$y'' = f(x, y, y'), \quad y(a) = \alpha, \quad y(b) = \beta \quad \longrightarrow \quad y'' = f, \quad y(a) = \alpha, \quad y'(a) = s \quad (\text{guess the slope } s).$$

The Boundary-Mismatch Function

$$F(s) = y(b; s) - \beta, \quad \text{the shooting method seeks a root } F(s) = 0 \text{ so the far BC is met.}$$

Linear BVP — Two Shots and Interpolate

$$F \text{ affine in } s \Rightarrow s^* = s_0 + (s_1 - s_0) \frac{\beta - y(b; s_0)}{y(b; s_1) - y(b; s_0)} \quad (\text{exact after two shots}).$$

Method — Shoot a Boundary Value Problem

1. Convert the BVP to an IVP by replacing the far condition $y(b) = \beta$ with a guessed slope $y'(a) = s$.
2. Integrate the IVP numerically (e.g. Runge–Kutta) from a to b and read off the overshoot/undershoot $F(s) = y(b; s) - \beta$.
3. Adjust s to drive $F(s) \rightarrow 0$: bisection or Newton in general; for a *linear* BVP, two shots fix the affine F exactly.
4. Interpolate to the exact slope s^* and integrate once more to produce the solution that meets both boundary conditions.

Worked Example — Two Shots on a Linear BVP

Solve $y'' = y$ on $[0, 1]$ with $y(0) = 0$, $y(1) = 1$ by shooting (the problem is linear, so two shots suffice).

Recast: $y'' = y$, $y(0) = 0$, $y'(0) = s$; the exact IVP solution is $y(x; s) = s \sinh x$.

Shot $s_0 = 0$: $y(1; 0) = 0 \Rightarrow F(0) = 0 - 1 = -1$.

Shot $s_1 = 1$: $y(1; 1) = \sinh 1 \approx 1.1752 \Rightarrow F(1) = 1.1752 - 1 = 0.1752$.

Interpolate: $s^* = 0 + (1 - 0) \frac{1 - 0}{1.1752 - 0} = \frac{1}{\sinh 1} \approx 0.8509$.

Final shot: $y(x) = \frac{\sinh x}{\sinh 1}$, which satisfies $y(0) = 0$, $y(1) = 1$ exactly.

Key Takeaway

The shooting method trades a hard BVP for a familiar IVP by **guessing the missing initial slope** and then correcting it until the far boundary condition is hit. Root-finding on the mismatch $F(s) = y(b; s) - \beta$ does the correcting — and because F is affine for a linear BVP, **two shots and a straight-line interpolation** land it exactly, all while reusing standard Runge–Kutta machinery.

Common Pitfall

The clean two-shot interpolation is a **linear-problem privilege**. For a nonlinear BVP $F(s)$ is curved, so two shots do not suffice — you must iterate (Newton/bisection), and the IVP can be **unstable**, amplifying small slope errors over $[a, b]$ and ruining the far value.

17.5 Orthogonal Functions

The inner product $\langle f, g \rangle = \int_a^b f g dx$; orthogonality as a vanishing inner product; the sine orthogonality integral (0 for $m \neq n$, $L/2$ for $m = n$); the functional basis and the generalized Fourier coefficient $c_n = \langle f, y_n \rangle / \langle y_n, y_n \rangle$.

Inner Product and Orthogonality

$$\langle f, g \rangle = \int_a^b f(x) g(x) dx; \quad f \perp g \iff \langle f, g \rangle = 0 \text{ (weighted: } \langle f, g \rangle_w = \int_a^b f g w dx \text{)}.$$

The Sine Orthogonality Integral

$$\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \begin{cases} 0, & m \neq n, \\ \frac{L}{2}, & m = n. \end{cases}$$

Functional Basis — Generalized Fourier Expansion

$$f(x) = \sum_{n=1}^{\infty} c_n y_n(x), \quad c_n = \frac{\langle f, y_n \rangle_w}{\langle y_n, y_n \rangle_w} = \frac{\int_a^b f y_n w dx}{\int_a^b y_n^2 w dx}.$$

Method — Expand a Function in an Eigenfunction Series

1. Identify the orthogonal eigenfunctions $\{y_n\}$ and the weight w for the underlying BVP (for a string, $y_n = \sin(n\pi x/L)$, $w = 1$).
2. Write the target expansion $f(x) = \sum_n c_n y_n(x)$ and isolate a single coefficient by taking the inner product of both sides with y_m .
3. Apply orthogonality: every term with $n \neq m$ vanishes, collapsing the infinite sum to one surviving term $c_m \langle y_m, y_m \rangle$.
4. Solve $c_m = \langle f, y_m \rangle / \langle y_m, y_m \rangle$ by evaluating the numerator and the (known) normalization integral in full.

Worked Example — Vibrating String: Eigenfunction Expansion of $f(x) = x$

A string of length L is fixed at both ends, so its spatial modes solve the eigenvalue BVP $X'' + \lambda X = 0$, $X(0) = X(L) = 0$. (1) Set up the operator and BVP, (2) find the eigenvalues and eigenfunctions, (3) build the coefficient formula from orthogonality, and (4) expand the initial profile $f(x) = x$ on $[0, L]$ as $f(x) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{L}$, computing every integral in full.

Step 1 — Operator and BVP. $\mathcal{L}[X] = -X'' = \lambda X$, $X(0) = X(L) = 0$: a regular Sturm–Liouville problem.

Step 2 — Eigenvalues. Only $\lambda = \mu^2 > 0$ is nontrivial: $X = A \cos \mu x + B \sin \mu x$, $X(0) = A = 0$, $X(L) = B \sin \mu L = 0$.

$$\text{Eigenpairs: } \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

Step 3 — Coefficient formula. Take $\langle \cdot, \sin \frac{n\pi x}{L} \rangle$ of $f = \sum_m c_m \sin \frac{m\pi x}{L}$; orthogonality kills $m \neq n$: $c_n = \frac{2}{L} \int_0^L x \sin \frac{n\pi x}{L} dx$.

Denominator (normalization). $\int_0^L \sin^2 \frac{n\pi x}{L} dx = \int_0^L \frac{1 - \cos \frac{2n\pi x}{L}}{2} dx = \frac{1}{2} \left[x - \frac{L}{2n\pi} \sin \frac{2n\pi x}{L} \right]_0^L = \frac{1}{2} L$

Numerator — set up by parts. $I_n = \int_0^L x \sin \frac{n\pi x}{L} dx$; $u = x$, $dv = \sin \frac{n\pi x}{L} dx \Rightarrow du = dx$, $v = -\frac{L}{n\pi} \cos \frac{n\pi x}{L}$.

$$I_n = \left[-\frac{Lx}{n\pi} \cos \frac{n\pi x}{L} \right]_0^L + \frac{L}{n\pi} \int_0^L \cos \frac{n\pi x}{L} dx \quad (\text{integration by parts}).$$

$$\text{Boundary term: } \left[-\frac{Lx}{n\pi} \cos \frac{n\pi x}{L} \right]_0^L = -\frac{L^2}{n\pi} \cos n\pi - 0 = -\frac{L^2}{n\pi} (-1)^n = (-1)^{n+1} \frac{L^2}{n\pi}.$$

$$\text{Remaining integral: } \frac{L}{n\pi} \int_0^L \cos \frac{n\pi x}{L} dx = \frac{L}{n\pi} \left[\frac{L}{n\pi} \sin \frac{n\pi x}{L} \right]_0^L = \frac{L^2}{n^2 \pi^2} (\sin n\pi - 0) = 0.$$

$$\text{Combine: } I_n = (-1)^{n+1} \frac{L^2}{n\pi} + 0 = (-1)^{n+1} \frac{L^2}{n\pi}.$$

$$\text{Coefficient: } c_n = \frac{I_n}{L/2} = (-1)^{n+1} \frac{L^2}{n\pi} \cdot \frac{2}{L} = (-1)^{n+1} \frac{2L}{n\pi}.$$

Step 4 — Expansion. $x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2L}{n\pi} \sin \frac{n\pi x}{L} = \frac{2L}{\pi} \left(\sin \frac{\pi x}{L} - \frac{1}{2} \sin \frac{2\pi x}{L} + \frac{1}{3} \sin \frac{3\pi x}{L} - \dots \right).$

Key Takeaway

Orthogonality is the engine of every eigenfunction expansion. Because $\langle y_n, y_m \rangle = 0$ for $n \neq m$, taking the inner product of $f = \sum c_m y_m$ with a single y_n **annihilates the entire sum but one term**, so each coefficient $c_n = \langle f, y_n \rangle / \langle y_n, y_n \rangle$ is computed *independently* by one integral — never by solving an infinite coupled system.

Common Pitfall

Do not forget to **divide by the normalization** $\langle y_n, y_n \rangle$ — here the $L/2$, not 1. Writing $c_n = \int_0^L f \sin(n\pi x/L) dx$ alone is off by the factor $2/L$. And when a **weight** $w(x) \neq 1$ is present, both integrals must carry it: $c_n = \int f y_n w dx / \int y_n^2 w dx$.

17.6 Sturm–Liouville Theory

The Sturm–Liouville form $(p y')' + (q + \lambda w)y = 0$; regular S–L problems and their guarantees (real eigenvalues, $\lambda_n \rightarrow \infty$, w -orthogonal eigenfunctions, completeness); the Lagrange-identity proof of orthogonality; converting an equation to S–L form.

The Sturm–Liouville Form

$$(p(x) y')' + (q(x) + \lambda w(x))y = 0, \quad p, w > 0 \text{ on } [a, b], \quad \text{with homogeneous (separated) BCs.}$$

Guarantees of a Regular S–L Problem

(i) eigenvalues λ_n real, $\lambda_1 < \lambda_2 < \dots \rightarrow \infty$; (ii) $\langle y_n, y_m \rangle_w = 0$ ($n \neq m$); (iii) $\{y_n\}$ complete.

Putting an Equation in S–L Form

$$a_2 y'' + a_1 y' + a_0 y + \lambda y = 0 \xrightarrow{\mu = \frac{1}{a_2} e^{\int a_1/a_2 dx}} (p y')' + (q + \lambda w)y = 0, \quad p = \mu a_2, \quad w = \mu.$$

Method — Prove Eigenfunction Orthogonality

1. Take two eigenpairs (λ_n, y_n) and (λ_m, y_m) , $\lambda_n \neq \lambda_m$: $(p y_n')' + (q + \lambda_n w)y_n = 0$ and $(p y_m')' + (q + \lambda_m w)y_m = 0$.
2. Multiply the first by y_m and the second by y_n , then subtract; the q -terms cancel, leaving $(\lambda_n - \lambda_m)w y_n y_m = y_n (p y_m')' - y_m (p y_n')'$.
3. Recognize the right side as an exact derivative (Lagrange's identity): $y_n (p y_m')' - y_m (p y_n')' = \frac{d}{dx} [p (y_n y_m' - y_m y_n')]$.
4. Integrate over $[a, b]$: $(\lambda_n - \lambda_m) \int_a^b w y_n y_m dx = [p (y_n y_m' - y_m y_n')]_a^b$.
5. The separated homogeneous BCs make the bracket vanish at both ends; since $\lambda_n \neq \lambda_m$, conclude $\int_a^b y_n(x) y_m(x) w(x) dx = 0$.

Worked Example — The String Problem as a Sturm–Liouville Problem

Cast $X'' + \lambda X = 0$, $X(0) = X(L) = 0$ in Sturm–Liouville form, name p, q, w , and confirm the eigenfunctions $\sin(n\pi x/L)$ are w -orthogonal.

Match the form: $(pX')' + (q + \lambda w)X = 0$ with $X'' + \lambda X = 0 \Rightarrow p(x) = 1$, $q(x) = 0$, $w(x) = 1$.

Regularity: $p = 1 > 0$, $w = 1 > 0$ on $[0, L]$ with separated BCs $X(0) = X(L) = 0 \Rightarrow$ a regular S–L problem.

Boundary term vanishes: $[p(y_n y'_m - y_m y'_n)]_0^L = 0$ since $\sin \frac{n\pi x}{L} = 0$ at $x = 0, L$.

Orthogonality confirmed: $\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} (1) dx = 0$ ($n \neq m$), $= \frac{L}{2}$ ($n = m$).

Conclusion: $\{\sin(n\pi x/L)\}$ is the complete, w -orthogonal eigenbasis predicted by S–L theory.

Key Takeaway

Sturm–Liouville theory is the structural payoff of the unit: *any* second-order linear eigenvalue problem can be written as $(py')' + (q + \lambda w)y = 0$, and once it is **regular**, three gifts follow automatically — the eigenvalues are **real and unbounded**, the eigenfunctions are **orthogonal with respect to the weight w** , and they form a **complete basis**. That completeness is exactly what licenses the generalized Fourier expansion $f = \sum c_n y_n$.

Common Pitfall

Eigenfunctions are orthogonal with respect to the **weight $w(x)$** , **not plain dx** — whenever $w \neq 1$, the inner product and every coefficient integral must carry it. Also, the orthogonality proof leans on the boundary bracket $[p(y_n y'_m - y_m y'_n)]_a^b$ vanishing; this requires **genuinely separated homogeneous BCs**, so the guarantees can fail for a problem that is not a regular S–L problem.

Practice Problems

Work the following ten problems in order. They climb from telling an IVP from a BVP and solving simple two-point problems, through the full three-case eigenvalue analysis and the shooting method, up to the orthogonality integral, a complete eigenfunction expansion with every coefficient integral expanded, and the Sturm–Liouville orthogonality proof. Complete solutions for all problems follow.

1. **(IVP vs. BVP)** State what distinguishes a boundary value problem from an initial value problem, and identify which of $y(0) = 1, y'(0) = 2$ versus $y(0) = 1, y(2) = 5$ is which.
2. **(Simple Dirichlet BVP)** Solve $y'' = 0$ on $[0, 2]$ with $y(0) = 1, y(2) = 5$.
3. **(Nonhomogeneous BVP)** Solve $y'' = 2$ on $[0, 1]$ with $y(0) = 0, y(1) = 0$.
4. **(Infinitely many solutions)** Show that $y'' + y = 0$ with $y(0) = 0, y(\pi) = 0$ has infinitely many solutions, and give their form.
5. **(Eigenvalues, three cases)** For $y'' + \lambda y = 0, y(0) = y(L) = 0$, carry out the $\lambda < 0, \lambda = 0, \lambda > 0$ analysis and state λ_n and y_n .
6. **(Neumann spectrum)** Find the eigenvalues and eigenfunctions of $y'' + \lambda y = 0$ with $y'(0) = y'(L) = 0$.
7. **(Shooting method)** Solve $y'' = y$ on $[0, 1], y(0) = 0, y(1) = 1$ by two shots and linear interpolation.
8. **(Orthogonality integral)** Evaluate $\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx$ for $m \neq n$ and for $m = n$.
9. **(Eigenfunction expansion)** Expand $f(x) = x$ on $[0, L]$ in the sine series $\sum_{n \geq 1} c_n \sin \frac{n\pi x}{L}$, computing c_n in full.
10. **(Sturm–Liouville capstone)** Write $X'' + \lambda X = 0, X(0) = X(L) = 0$ in S–L form (name p, q, w), then prove $\int_a^b y_n y_m w dx = 0$ for $\lambda_n \neq \lambda_m$.

Complete Solutions

Solution 1. A BVP imposes its side conditions at *two different points*, whereas an IVP stacks all of them at a single point. Hence

$$y(0) = 1, \quad y'(0) = 2 \text{ (both at } x = 0) \Rightarrow \text{IVP}; \quad y(0) = 1, \quad y(2) = 5 \text{ (at } x = 0, 2) \Rightarrow \text{BVP}.$$

Unlike the IVP, the BVP carries no automatic existence–uniqueness guarantee.

Solution 2. The general solution of $y'' = 0$ is $y = c_1 + c_2x$. Imposing the endpoints,

$$y(0) = c_1 = 1; \quad y(2) = 1 + 2c_2 = 5 \Rightarrow c_2 = 2.$$

Unique solution: $y(x) = 1 + 2x$.

Solution 3. Integrate twice: $y = x^2 + c_2x + c_1$. Then

$$y(0) = c_1 = 0; \quad y(1) = 1 + c_2 = 0 \Rightarrow c_2 = -1.$$

Unique solution: $y(x) = x^2 - x$.

Solution 4. The general solution is $y = A \cos x + B \sin x$. The conditions give

$$y(0) = A = 0; \quad y(\pi) = A \cos \pi + B \sin \pi = -A = 0,$$

which is satisfied for *every* B . Thus $y = B \sin x$ is a solution for all B — infinitely many (here $\lambda = 1$ is an eigenvalue).

Solution 5. For $\lambda < 0$ and $\lambda = 0$ the BCs force $y \equiv 0$. For $\lambda = \mu^2 > 0$, $y = A \cos \mu x + B \sin \mu x$ with $y(0) = A = 0$, so $y = B \sin \mu x$ and $y(L) = B \sin \mu L = 0$:

$$\sin \mu L = 0 \Rightarrow \mu L = n\pi \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad y_n = \sin \frac{n\pi x}{L}, \quad n = 1, 2, 3, \dots$$

Solution 6. With $y = A \cos \mu x + B \sin \mu x$, $y' = \mu(-A \sin \mu x + B \cos \mu x)$. Then $y'(0) = \mu B = 0 \Rightarrow B = 0$, and $y'(L) = -\mu A \sin \mu L = 0$:

$$\sin \mu L = 0 \Rightarrow \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad y_n = \cos \frac{n\pi x}{L}, \quad n = 0, 1, 2, \dots$$

The Neumann case keeps $n = 0$: $\lambda_0 = 0$ with constant eigenfunction $y_0 = 1$.

Solution 7. Recast as $y'' = y$, $y(0) = 0$, $y'(0) = s$, with exact IVP solution $y(x; s) = s \sinh x$. Two shots:

$$F(0) = 0 - 1 = -1; \quad F(1) = \sinh 1 - 1 \approx 0.1752; \quad s^* = \frac{1}{\sinh 1} \approx 0.8509.$$

Final solution: $y(x) = \frac{\sinh x}{\sinh 1}$, exact since the problem is linear.

Solution 8. Use $\sin \alpha \sin \beta = \frac{1}{2}[\cos(\alpha - \beta) - \cos(\alpha + \beta)]$. For $m \neq n$ both cosine integrals vanish over $[0, L]$, giving 0. For $m = n$,

$$\int_0^L \sin^2 \frac{n\pi x}{L} dx = \int_0^L \frac{1 - \cos \frac{2n\pi x}{L}}{2} dx = \frac{L}{2}.$$

So the integral is 0 for $m \neq n$ and $L/2$ for $m = n$.

Solution 9. With weight $w = 1$, $c_n = (\int_0^L x \sin \frac{n\pi x}{L} dx) / (\int_0^L \sin^2 \frac{n\pi x}{L} dx)$. The denominator is $L/2$; integrating the numerator by parts,

$$\int_0^L x \sin \frac{n\pi x}{L} dx = \left[-\frac{Lx}{n\pi} \cos \frac{n\pi x}{L} \right]_0^L + \frac{L}{n\pi} \int_0^L \cos \frac{n\pi x}{L} dx = (-1)^{n+1} \frac{L^2}{n\pi} + 0.$$

$$c_n = \frac{(-1)^{n+1} L^2 / (n\pi)}{L/2} = (-1)^{n+1} \frac{2L}{n\pi} \Rightarrow x = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2L}{n\pi} \sin \frac{n\pi x}{L}.$$

Solution 10. Matching $(pX')' + (q + \lambda w)X = 0$ to $X'' + \lambda X = 0$ gives $p = 1$, $q = 0$, $w = 1$ (regular, since $p, w > 0$ with separated BCs). For the proof, multiply $(py'_n)' + \lambda_n w y_n = 0$ by y_m , the m -equation by y_n , subtract, and integrate:

$$(\lambda_n - \lambda_m) \int_a^b w y_n y_m dx = [p(y_n y'_m - y_m y'_n)]_a^b = 0,$$

where the bracket vanishes by the homogeneous BCs. Since $\lambda_n \neq \lambda_m$, $\int_a^b y_n y_m w dx = 0$.

Unit Key Takeaway

One thread ties the unit together: **boundary conditions create a spectrum, and orthogonality turns that spectrum into a basis**. Splitting the side data across two endpoints can leave a linear problem with no solution, one, or infinitely many — the last case is where the **eigenvalues** $\lambda_n = (n\pi/L)^2$ and **eigenfunctions** $\sin(n\pi x/L)$ live. Those eigenfunctions are **orthogonal with respect to a weight** w , so any reasonable f expands as $f = \sum c_n y_n$ with each $c_n = \langle f, y_n \rangle / \langle y_n, y_n \rangle$ a single integral. The **Sturm–Liouville form** $(py')' + (q + \lambda w)y = 0$ is the master frame that guarantees real eigenvalues, weighted orthogonality, and a complete functional basis.